

BSE Group Assignments: Task Sheet

Mathematics for Machine Learning, Formative 2 PCA, Cohort 2, Team 15

Team Members

Team Members	Role / Responsibility	Email
Teta Butera Nelly	Data handling, PCA code, plots, benchmarking, GitHub repo setup	t.nelly@alustudent.com
Mutoni Keira	Written answers, Colab check, GitHub README and data dictionary	m.keira@alustudent.com

Task Allocation and Tracker

Task Allocated	Assigned Member	Deadline	Completion Status	Reviewed by Team?	Comments
Chose the dataset: Our World in Data energy data for 54 African countries (2000-2022), checked against the brief (African, 7+ columns, real missing values, text columns)	Teta Butera Nelly	2026-09-30	Completed	Yes	944 real blank cells in the raw file, 19 features in the end.
Data handling: found the missing cells, filled them from each country's own years (sub-region median for the four countries with no GDP), encoded the text columns and log-transformed the skewed ones	Teta Butera Nelly	2026-09-30	Completed	Yes	Every step is explained in the notebook.
PCA from scratch in numpy: standardization, covariance matrix, eigendecomposition, sorting and projection (Steps 1-7), with checks that each result is correct	Teta Butera Nelly	2026-09-30	Completed	Yes	Eigenvalues sum to exactly 19, eigenvectors are orthonormal.
Explained variance and choosing the number of components from a variance threshold, with scree and cumulative variance plots (Task 2 code)	Teta Butera Nelly	2026-09-30	Completed	Yes	7 components keep 90% of the variance.
Before and after PCA plots, component loadings and a chart of the information lost per feature (Step 8)	Teta Butera Nelly	2026-09-30	Completed	Yes	
Performance benchmarking for large data: loops vs vectorized, float32 vs float64, chunked covariance and SVD, up to 1,000,000 rows (Task 3)	Teta Butera Nelly	2026-09-30	Completed	Yes	

Wrote the covariance explanation and the three interpretation answers (before/after plot, choice of components and trade-offs, information lost)	Mutoni Keira	2026-09-30	Completed	Yes	In the notebook and in the Google Doc.
Ran the notebook in Google Colab and checked every output, then post a short verification note on GitHub	Mutoni Keira	2026-09-30	In progress	No	Re-ran it in Colab with no errors. The GitHub note is still to post.
GitHub: accepted the invite, worked on a branch and opened a pull request for the README and data dictionary	Mutoni Keira	2026-09-30	Completed	Yes	Pull request #1 is merged.
GitHub: added the how-to-run-in-Colab section and a results summary to the README	Mutoni Keira	2026-09-30	Completed	Yes	
GitHub: wrote data/DATA_DICTIONARY.md covering the 15 features and the sub_region column	Mutoni Keira	2026-09-30	Completed	Yes	
Final steps: download the Google Doc as a PDF, merge it with the task sheet PDF, final check and submit on Canvas	Teta Butera Nelly and Mutoni Keira	2026-09-30	In progress	No	Last step.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
2026-09-28	Work distribution	Teta Butera Nelly	Teta Butera Nelly and Mutoni Keira	Went through the PCA brief and the rubric together. Agreed who does what: Nelly takes the data, the PCA code, the plots and the benchmarking. Mutoni takes the written answers, the Colab check and the GitHub README and data dictionary.	Nelly to find a dataset and build the notebook. Mutoni to start on her parts once the notebook is ready.
2026-09-30	Execution	Teta Butera Nelly	Teta Butera Nelly and Mutoni Keira	Notebook, GitHub repo and report are done. Mutoni added her written answers, the README and the data dictionary, and ran the notebook in Colab. Went over what's left: the task sheet, the final PDF and submitting.	Fill in the task sheet, download the Doc as a PDF, merge it with the task sheet PDF and submit on Canvas.